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THE VEGETATION OF A TROPICAL AFRICAN LAKE: CLASSIFICATION AND ORDINATION OF THE VEGETATION OF LAKE CHILWA (MALAWI)

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INTRODUCTION

Aquatic plant communities cover large areas of Africa and are important in relation to the total plant cover of tropical Africa (Walter 1971). Swamp communities are particularly common in East and Central Africa, occupying extensive areas of the Sudan (Sculthorpe 1967), Uganda (Carter 1955; Beadle & Lind 1960; Walter 1971), South Katanga and the Congo Basin (Marlier 1973), North-East and Western Zambia, Malawi, and the Okavango River system in Botswana (Wild 1961). In spite of this, there are few published data concerning the vegetation of African water bodies. Valuable qualitative background data are given by Eggeling (1935), Lind (1956), Noel (1959), Cook (1968), Mitchell (1969), and Vesey-Fitzgerald (1970), but there appears to be no published quantitative account of African swamp vegetation. This lack of data is at least partly because most swamp areas are difficult to reach and often almost impossible to penetrate. This paper presents a quantitative account of the vegetation of an African swamp, and relates the vegetation to measured environmental factors. The swamp studied was that surrounding Lake Chilwa in Malawi (Fig. 1). This lake has been the subject in recent years of a number of biological studies, covering all trophic levels (e.g. Moss & Moss 1969; McLachlan & McLachlan 1969; Morgan & Kalk 1970; Morgan 1972; McLachlan *et al.* 1972; Howard-Williams 1972).

THE STUDY AREA

Comprehensive accounts of Lake Chilwa are given in the studies listed above, so only a brief account is given here.

Physiography

Lake Chilwa lies between latitudes 15° 0' and 15° 30' S and longitudes 35° 30' and 36° 0' E, and is situated at *c.* 620 m altitude in a shallow tectonic depression 100 km south-east of Lake Malawi. The lake is endorheic and has an open water surface area of 700 km², in a drainage basin with a catchment area of 7000 km² (Fig. 1). The lake is extremely shallow (maximum depth 500 cm) and is completely surrounded by large tracts of *Typha domingensis* swamp in the north and east, and smaller areas in the west and south (Fig. 1). The area occupied by swamp is approximately equal to the area of

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open water, and an additional 500 km² of marsh and grassland mainly in the north and south are flooded for short periods each year. These areas lie within the 3 m raised terrace (Fig. 1) which delineates the study area.

Much of the bedrock of the Chilwa basin is covered by alluvium about 50 cm thick, consisting mainly of dark clay and silt with some sand (Bloomfield 1965). A raised beach at approximately 16 m above the present dry season lake level appears to indicate a stable phase in the lake's history when sandbars and spits developed and when the lake was part of an exorheic system, drained by the Lujenda River in the north, and connected with the present Lake Chiuta (Fig. 1) (see Garson 1960). The lake is at present separated from Lake Chiuta by a bar of wind-blown sand only 15 m high. The length of time that the lake has been endorheic is still a matter of controversy.

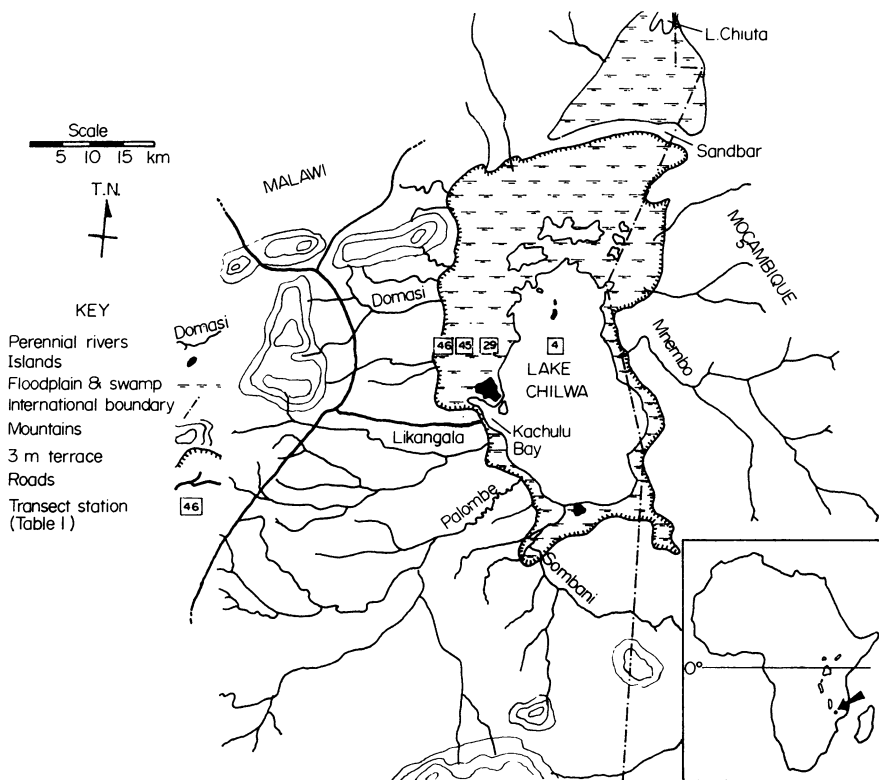


FIG. 1. The Lake Chilwa basin, showing inflowing rivers, the sandbar which has blocked off the outflow, and the large swamp and grassland areas surrounding the open lake. The numbers 46, 45, 29 and 4 refer to stand localities referred to in Table 1. Inset: location of Lake Chilwa.

Climate

Lake Chilwa has two distinct climatic seasons, a cool dry season from April to November, and a warmer rainy season from December to March. Air temperatures during the day are fairly high throughout the year with maxima of over 30° C for seven months (Morgan & Kalk 1970). The average annual rainfall for the years 1961–1971 was 960 mm. Most of the area bounded by the 3 m terrace is flooded during the rainy season, and over

one-third of the area dries out during the dry season. The average value for solar radiation for the period 1968–1971 was $1888 \times 10^3 \text{ kcal m}^{-2} \text{ yr}^{-1}$ ($7900 \times 10^3 \text{ kJ m}^{-2} \text{ yr}^{-1}$) with maxima in March and October. There are between 5.4 and 9.8 hours of sunshine per day, the lowest figures being recorded at the height of the rainy season (figures were taken from values recorded at Monkey Bay, Lake Malawi, by Pike & Rimmington 1965). The strongest winds of the year blow from the south or south-east during the dry season. Rainy season winds are generally northerly, although short-lived local winds can blow from any direction.

Hydrology

Only five rivers in the radial drainage pattern (Fig. 1) are perennial. The seasonal changes between wet and dry periods, together with high evaporation rates, cause large annual fluctuations in the lake level, about half the lake water being replaced each year (Moss & Moss 1969). The vegetation is affected both by the resulting variations in the

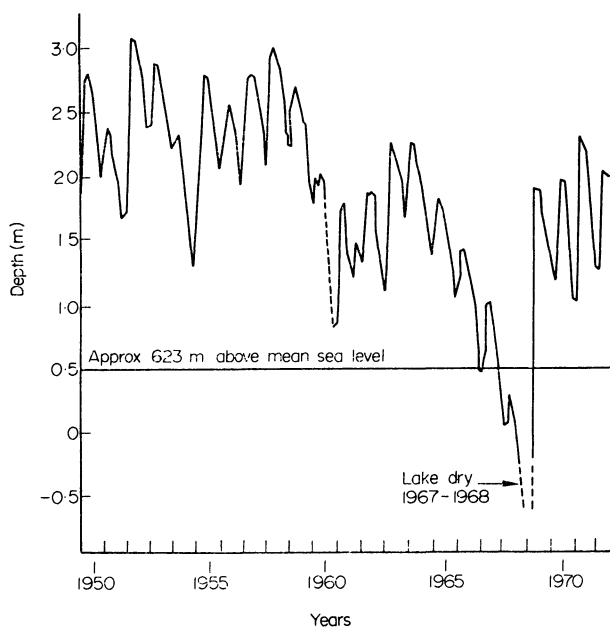


FIG. 2. Annual fluctuations in water level since 1949. Note the dry period in 1968, and also the extent of the annual fluctuations, often more than one-third of the total depth of the lake.

levels of dissolved salts and by the seasonal drying and wetting of 600 km^2 of marginal land around the shallow-sloping shoreline. This annual hydrological cycle is important not only in determining the composition of the vegetation, but also in determining the characteristics of the soil in any particular area (Howard-Williams 1972). The amplitude of water level fluctuations from 1949 to 1972 is shown in Fig. 2. The recent dry period shown here has been discussed fully by Morgan (1968), Moss & Moss (1969), McLachlan & McLachlan (1969) and Morgan & Kalk (1970). This period had an effect on the vegetation, of which brief mention is made later.

Physical and chemical characteristics of soils and waters

The physico-chemical characteristics of the open lake have been described in detail by McLachlan *et al.* (1972). Lake Chilwa is extremely turbid (Secchi disc disappears at 10 to 12 cm), and alkaline (pH between 7.5 and 9.3) with a relatively high ionic content (conductivity averaging about 2000 $\mu\text{mho cm}^{-1}$). The swamp waters undergo marked seasonal variations in both water chemistry and physical properties (Howard-Williams 1972).

The soils of the Lake Chilwa basin are generally alkaline clays with a high exchange capacity. However, sandy soils occupy fairly extensive areas (McLachlan *et al.* 1972) and a land-to-lake gradient in soil physical and chemical characteristics was shown by Howard-Williams (1972). Table 1 gives selected soil and water characteristics from four stations in a transect from land to lake, which illustrate this gradient. The station positions are shown in Fig. 1.

Table 1. *Soil and water characteristics on a land-to-lake transect in Lake Chilwa (the four stations are shown in Fig. 1)*

Station number Description of station	46 Land	45 Swamp	29 Swamp	4 Lake
Soil properties				
pH	4.8	6.5	8.0	9.3
Conductivity ($\mu\text{mho cm}^{-1}$)	300	400	3000	2500
Cation exchange capacity (m-equiv./100 g)	5	35	30	26
Organic matter (% dry weight)	1.7	5.9	5.5	2.7
Clay (%)	8	43	71	73
Silt (%)	3	16	8	8
Sand (%)	91	46	14	1
Water properties				
pH	—	7.1	7.7	9.1
Conductivity ($\mu\text{mho cm}^{-1}$)	—	300	2000	2000
Clarity (% transmission at 430 nm)	—	94	87	43

Soil and water pH and conductivity increase from land to lake. Percentage sand decreases, and percentage clay increases. Cation exchange capacity and organic matter increase in the swamp and decrease into the lake. Water clarity is much lower in the lake than in the swamp.

Human influences

Drummond (1902), who visited Lake Chilwa in 1889, comments, 'owing to the ravages of the slaver the people of Shirwa are few, scattered and poor . . .'. Today, population densities on the landward side of the marshes vary around 100 km^{-2} (Agnew 1970). The large herds of wild animals which existed around the lake formerly (Drummond 1902) have disappeared, the remnants being a few reedbuck (*Redunca arundium* Boddaert) and *Hippopotamus* (*H. amphibius* L.). Cattle have replaced grazing game animals. About one-third of the marginal area of Lake Chilwa is burnt each year during the dry season. The fires are started deliberately, and in a strong wind even swamp vegetation standing in water is burnt. Several rice-growing schemes have recently been started on the less alkaline of the floodplain soils (Kalk 1970). Of particular importance to this study are canoe channels cut through the swamp from lake to land by fishermen. These channels are normally just wide enough for the passage of a dugout canoe and are up to

11 km long. They rapidly become overgrown when not in use but do allow access to the otherwise impenetrable central regions of the swamps. At intervals along the canoe channels the fishermen cut patches of *Typha* up to 30 m in diameter, heap the cut material into a mound in the centre of the clearing, and weave a temporary reed hut on the mound. These mounds are often abandoned at the end of the dry season.

METHODS

Sampling procedure

During the rainy season, the marginal areas of Lake Chilwa are largely inaccessible, and sampling of vegetation was therefore carried out in the dry season. This had the effect of reducing the number of species encountered, as only the perennial species were alive in the dry season. As annuals are relatively infrequent and small (Howard-Williams 1973), their inclusion would not greatly affect the classification or ordination of the vegetation.

In order to include as much of the variation in the study area as possible, sampling localities selected on the basis of environmental differences alone were sited over a large part of the lake basin to include any spatial variation, for example between northern and southern floodplains. Stands were selected in each sampling locality according to the criteria of Walker & Coupland (1968, 1970). The stands had to be: (i) dominated by native species; (ii) visually homogeneous with regard to vegetation, i.e. with no obvious visual discontinuities; (iii) at least 100 m² in area.

A systematic sampling design was used within each stand, the position of the first sample being selected at random, either by walking a random number of paces or by pushing a boat a random number of pole strokes (Howard-Williams 1971). Vegetation and selected environmental factors were measured at 73 stands.

Frequency sampling was used for the vegetation, employing three-sided open quadrats. In each stand 50 quadrats were recorded, in five transects of ten quadrats each. An initial survey was made on ten stands to determine the appropriate quadrat size (Hyder *et al.* 1963; Rice 1967). Quadrat sizes of 7.5, 15, 25, 35, 40 and 50 cm square were tried. The 50 cm square size, which gave a frequency of about 80% for the commonest species, was chosen (cf. Curtis & McIntosh 1950). Vegetation in deep water stands was sampled from a boat as described by Howard-Williams (1971). Although the confidence limits for 50 samples are wide, the differences between stands were expected to be even larger. Walker & Coupland (1970) found that the differences in species composition in simple wetland stands in Canada allowed the use of only 10 to 30 quadrats per stand.

A presence list of species was also made for each stand, which for this purpose was delineated by the two outer transects. Frequency values of 1% were given to all species included in the presence list, but not included in the quadrat sampling. Species present in only one stand, or in less than four stands and with a frequency of less than 5% in all of these, were omitted, to reduce the data matrix to a manageable size.

At each stand where surface water was present, a water sample was collected. A corer was used to obtain a sediment sample 20 to 30 cm deep. This was mixed and air-dried within 10 hours of collection. The light intensity at water level was recorded on an EEL portable light meter and expressed as a percentage of the value above the vegetation. Water depth was measured, and degree of dryness estimated using a scale based on the number of days between November 1970 and November 1971 that a stand was dry. The scale was as follows: 0, water present all the year; 1, less than 50 days dry; 2, 50 to

100 days dry; 3, 100 to 150 days dry; 4, 150 to 200 days dry; 5, 200 to 250 days dry; 6, 250 to 300 days dry; 7, dry for over 300 days per year. The scale value for each stand was calculated from the water level fluctuation during this period measured at monthly intervals, and by frequently noting the extent of the water surface around the lake at the different water depths.

The degree of disturbance at each site was estimated on the following scale: 1, undisturbed; 2, apparently undisturbed, but with signs of a type 6 area within 50 m; 3, within 5 m of a canoe channel or path; 4, visual signs of cutting, grazing or trampling; 5, at least 50% of the vegetation grazed or trampled; 6, all the vegetation grazed or trampled, or indications that the area had been cultivated or used for habitation within the previous year.

Exposure to wind and wave action was assessed for swamp and lake stands on the following scale: 1, always calm, over 50 m inside the vegetation; 2, between 10 and 50 m inside the vegetation or anywhere on the edge of the vegetation with a fetch of less than 1 km, or at the sheltered north end of Kachulu Bay (Fig. 1); 3, anywhere in the open lake, or up to 10 m inside the swamp fringe facing an area of open water with a fetch of 1 to 5 km; 4, anywhere in the open lake or less than 10 m inside the swamp fringe facing an area of open water with a fetch of more than 5 km.

As only surface soil samples were collected, it was thought that a layer of sand or gravel more than 30 cm below the surface would be undetected, and might have an important effect on the vegetation. The depth of fine sediments was therefore recorded by pushing a wooden rod slowly through the sediments until sand or gravel was felt. Such layers over 1 m below the surface were not recorded.

Soil and water analysis

Water clarity was measured as percentage transmission at 430 nm (Howard-Williams 1972). Conductivity was measured on a Dionic water tester, and pH using a glass electrode. Subsequent analysis was carried out on samples filtered through Whatman GF/C filter paper. The major cations and anions are strongly correlated with conductivity (McLachlan *et al.* 1972; Howard-Williams 1973) and are not dealt with in this paper. Nitrate-nitrogen was measured by the method of Müller & Wideman (1965), and orthophosphate using the method of Murphy & Riley (1958).

Soil conductivity was measured using a Dionic water tester on water extracted by centrifugation from a saturation paste of the sediment. pH was measured on a 1:2.5 suspension of sediment in 0.01 M calcium chloride solution. Percentage silt and clay were measured by the hydrometer method, and coarse and fine sand by sieving. Undecomposed organic matter often interfered with the sieving, so values do not always add up to 100%. Cation exchange capacity (C.E.C.) was determined by the calcium chloride-ammonium acetate method (Jackson 1958). Organic carbon was determined using the Walkley-Black method (Walkley 1947); the results are expressed as organic matter after conversion using the van Bemmelen factor (Jackson 1958). Phosphorus was measured by Bray's method (Bray & Kurtz 1945).

Analysis of the species data

There were many zero values in the species-stand matrix. Owing to the limitations imposed on multivariate analytical techniques by a large number of zero values in a species-stand matrix, the species data were subjected to the zero transformation pro-

cedure of Swan (1970). As all species data were in the form of percentage frequency, no further transformation was considered necessary.

The data were analysed in two ways. First, in order to classify the vegetation, they were subjected to a cluster analysis, based on a matrix of inter-stand correlation coefficients. The effect of zero transformation by Swan's method on the cluster analysis has been discussed by Walker (1974) who found that it has the effect of eliminating negative correlations, but provided for ecologically more informative results than the untransformed data. The cluster analysis was done as follows: an intercorrelation matrix between the stands was computed using the species values as variates. The species values of the pair of stands with the highest correlation coefficient were then averaged to form a new stand. The correlation coefficients between this stand and all the others were computed to form a new correlation matrix with one less stand than the original. As the process was repeated, the species values of new stands were determined as a weighted mean of the component stands, the weighting being based on the number of stands involved. The process was continued until all the stands had been combined into a single group. Secondly, to determine the nature and relative importance of the major environmental gradients causing variation in the vegetation, the data were subjected to a principal components analysis (PCA). Principal components analyses with both zero transformed and untransformed data showed clearly that a greater proportion of the total variance could be explained using the transformed data (Howard-Williams 1973). The analysis was based on a covariance matrix according to the procedure of Walker & Wehrhahn (1971). This produced, initially, species loadings on components, and then stand position values for each component, derived by summing for each stand the product of the normalized species loading and the measured value in each stand. In order to determine the correlations between particular species and the extracted components, the loadings were then adjusted by multiplying the normalized loadings by $\sqrt{\lambda/S^2x}$, where λ is the eigen value and S^2x is the total variance of the species concerned.

Austin & Noy-Meir (1971) showed that the direct interpretation of ordination axes in terms of environmental gradients is seldom possible owing to the inherent non-linearity of vegetation–environment relationships. They did, however, suggest three ways to reduce this problem. The first is by the examination of only a narrow range of vegetational and environmental gradients, the second by using methods which correct for non-linear relationships, e.g. by zero transformation, and the third by limiting the detailed interpretation of the ordination. All these suggestions have to a large extent been followed in this study.

The relative importance of each vegetation type in the Lake Chilwa basin was estimated by constructing a vegetation map on which the vegetation groupings derived from the cluster analysis were superimposed. The map was constructed after extensive ground and aerial surveys.

NOMENCLATURE AND TERMINOLOGY

Botanical nomenclature is that given by Exell & Wild (1960), Exell, Fernandes & Wild (1963), Simon (1971) and Drummond (1972). The plants were named in the Government Herbarium, Salisbury, Rhodesia. Authorities for species are given in Table 2 and Appendix 1.

Many of the temperate classifications of wetland areas, such as those of Curtis (1959) and Ungar (1965), are not easily applied in Africa where terms such as 'prairie' and

Table 2. Average frequency and presence values of the species in the vegetation types derived from the cluster analysis; the intermediate and disturbed vegetation types have been omitted as they consist of only one and two stands respectively. F, % frequency of the species in the stands in which they occurred; P, % of the stands in each vegetation type in which the species occurred

	Flood-plain grassland		Division A Neutral to acidic marsh		Alkaline marsh		Division B Swamp transition		Swamp	
	F	P	F	P	F	P	F	P	F	P
<i>Scirpus littoralis</i> Schrad.	—	—	—	—	—	—	—	—	52	9
<i>Paspalidium geminatum</i> (Forsk.) Stapf	—	—	—	—	—	—	—	—	23	48
<i>Pseudowolfia hyalina</i> (Del.) den Hartog & van der Plas	—	—	—	—	—	—	—	—	58	9
<i>Ceratophyllum demersum</i> L.	—	—	—	—	—	—	78	11	38	55
<i>Pistia stratiotes</i> L.	—	—	—	—	—	—	68	11	39	10
<i>Lemna perpusilla</i> Torr.	—	—	—	—	—	—	44	11	38	6
<i>Panicum repens</i> L.	—	—	—	—	—	—	62	89	32	15
<i>Utricularia inflexa</i> Forsk.	—	—	—	—	—	—	18	22	14	42
<i>Diplachne fusca</i> (L.) Beauv.	—	—	—	—	—	—	35	56	22	27
<i>Nymphaea caerulea</i> Savigny	—	—	—	—	—	—	17	22	35	36
<i>Cyperus laevigatus</i> L.	—	—	—	—	—	—	20	11	12	3
<i>Spirodela polyrhiza</i> (L.) Schleid.	—	—	—	—	—	—	28	11	2	6
<i>Utricularia reflexa</i> Oliv.	—	—	—	—	—	—	35	33	3	3
<i>Cyperus alopecuroides</i> Rottb.	—	—	—	—	—	—	31	22	49	6
<i>Alternanthera sessilis</i> (L.) DC.	—	—	—	—	—	—	5	22	—	—
<i>Typha domingensis</i> Pers.	—	—	35	13	47	40	95	33	75	61
<i>Ipomoea aquatica</i> Forsk.	—	—	16	38	5	40	35	44	20	33
<i>Vossia cuspidata</i> Griff.	—	—	52	38	50	80	31	44	31	42
<i>Cyperus articulatus</i> L.	—	—	56	25	—	—	23	56	2	6
<i>Phragmites mauritanus</i> Kunth	—	—	27	44	—	—	—	—	39	15
<i>Echinochloa pyramidalis</i> (Lam.) Hutch.	—	—	71	44	100	20	—	—	20	3
<i>Aeschynomene pfundii</i> Taub.	10	42	—	—	12	20	1	11	11	25
<i>Cynodon dactylon</i> (L.) Pers.	75	100	64	25	68	60	—	—	33	6
<i>Cyperus procerus</i> Rottb.	8	14	50	50	—	—	51	33	—	—
<i>Scirpus maritimus</i> L.	56	14	53	13	—	—	49	44	—	—
<i>Utricularia gibba</i> L.	—	—	13	17	—	—	34	11	—	—
<i>Wolffiopsis welwitschii</i> (Hegelm.) den Hartog & van der Plas	—	—	6	13	—	—	—	—	—	—
<i>Cyperus longus</i> L.	6	14	—	—	45	100	—	—	—	—
<i>Aeschynomene nilotica</i> Taub.	—	—	3	19	—	—	—	—	—	—
<i>Oryza sativa</i> L.	—	—	14	13	—	—	—	—	—	—
<i>Cyperus digitatus</i> Roxb.	—	—	20	13	—	—	—	—	—	—
<i>C. papyrus</i> L.	—	—	69	19	—	—	—	—	—	—
<i>Pycnus mundtii</i> Nees	—	—	30	6	4	60	—	—	—	—
<i>Polygonum limbatum</i> Meisn.	—	—	8	13	—	—	—	—	—	—
<i>Leersia hexandra</i> Swartz	42	14	82	94	82	20	—	—	—	—
<i>Commelina africana</i> L.	18	14	56	6	—	—	—	—	—	—
<i>Eragrostis gangetica</i> (Roxb.) Steud.	35	57	64	13	22	40	—	—	—	—
<i>Sporobolus pyramidalis</i> Beauv.	24	43	18	13	—	—	—	—	—	—
<i>Eriochloa borumensis</i> Stapf	24	14	2	20	—	—	—	—	—	—
<i>Hyparrhenia rufa</i> (Nees) Stapf	78	100	—	—	—	—	—	—	—	—
<i>Hygrophila auriculata</i> (Schumach.) Heine	39	43	—	—	—	—	—	—	—	—
<i>Chloris virgata</i> Swartz	22	28	—	—	—	—	—	—	—	—

'meadow' are not used. Vesey-Fitzgerald (1970) separates swamps from valley and floodplain grasslands in East Africa, and Greenway (1973) defines these terms with examples from East African vegetation. The vegetation terminology used in this paper is as follows: (i) swamp—vegetation usually dominated by reeds or other tall grasses, sedges or rushes in which the normal rainy season water level is above the soil surface; (ii) marsh—vegetation usually dominated by sedges, occurring on waterlogged soils where the rainy season water level is close to but never much above the soil surface, and in which the soil has a low organic content; (iii) floodplain grassland—a grass-dominated community occurring on soils well above the dry season water level, but which are flooded during the rainy season.

RESULTS

Classification

One immediate problem which arises in the interpretation of a cluster analysis is to determine the different levels at which distinct groups can be separated. If the logarithm of the cumulative number of stands is plotted against the clustering level, the curve rises sharply at levels where a relatively large number of stands separates out on the clustering diagram. Fig. 3 shows the cluster analysis dendrogram together with the curve for determining the levels of group separation. The groups of stands which separated out at the 0.325 level are called Divisions, and represent the major split in the vegetation. Stands in Division A represent shallow water vegetation in areas which are dry for much of the year, whilst those in Division B represent deep water vegetation in areas which are flooded for all or most of the year. Eight vegetation types can be distinguished at the 0.60 level of clustering. Within each vegetation type, except where the type is represented by a single stand, groups and in one case subgroups have been separated. Groups were separated at the 0.725 clustering level. The distinction between subgroups is arbitrary. Three of the vegetation types are represented by single stands. Two of these (A5 and B1) are on disturbed areas so have been combined. The seven resulting vegetation types are given below.

Symbol	Vegetation type	No. of stands
A1	Floodplain grassland	7
A2	Neutral to acidic marsh	16
A3	Intermediate	1
A4	Alkaline marsh	5
A5 + B1	Disturbed areas	2
B2	Swamp transition	9
B3	Swamp	33

In order to present a quantitative record of the vegetation in the stands during the sampling period, the percentage frequency of each species in the stands in which it occurred, together with the percentage presence of each species, was calculated for each vegetation type. The results are given in Table 2, which therefore gives an indication of how often a species is likely to be found in each vegetation type and, when it is found, its frequency. The species have been arranged to show the change in species composition of the vegetation types from grassland to swamp. Inspection of the table shows that the species fall immediately into three groups. Those at the top of the table occur only in

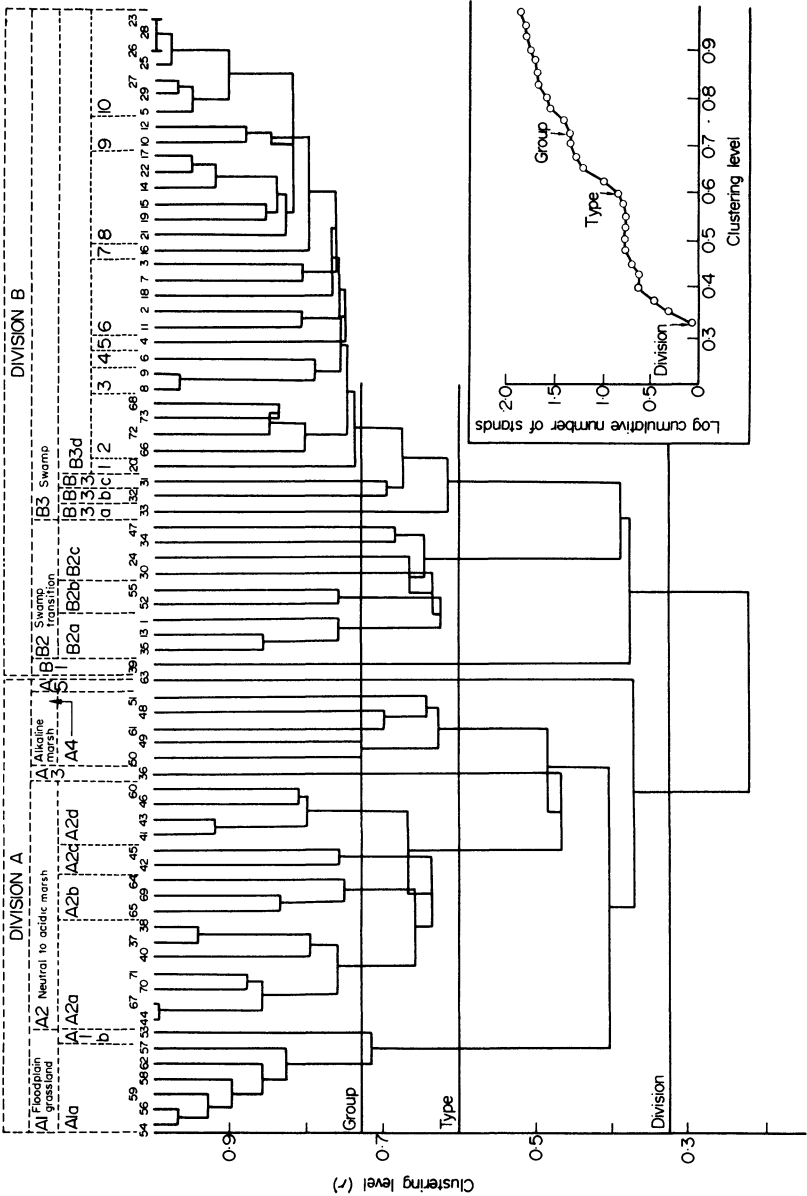


Fig. 3. Cluster analysis of the 73 stands. The basis of separation of stand clusters into divisions, types, and groups at different clustering levels is illustrated in the inset. The separation of group B3d into subgroups is based largely on field observation.

Division B and are deep water plants. Those in the centre of the table are species tolerant of a wide range of water depths, and those at the bottom occur only in drier areas represented by Division A.

The areas occupied by the vegetation types are shown in the vegetation map (Fig. 4). This was made using the classification derived from the cluster analysis.

Type A1: Floodplain grassland. This vegetation type can be equated partly to the valley grassland described by Vesey-Fitzgerald (1970). It occurs almost all the way round the edge of the lake basin except near perennial rivers, the most extensive areas being in the north. *Hyparrhenia rufa*, *Eragrostis gangetica*, *Sporobolus pyramidalis* and *Cynodon dactylon* are the commonest species. The stands are dry for about nine months each year, and all had a soil solution conductivity below $400 \mu\text{mho cm}^{-1}$.

Type A2: Neutral to acidic marsh. This is found mostly along the western shore of the lake and can be divided into four groups which are described separately. The stands in this vegetation type occurred in soils with a high silt content and a wide range of soil pH, from 7.0 to 4.5.

Group A2a is found near perennial rivers (Fig. 4) where the average soil pH was 5.4. The vegetation is an association of the tall grasses *Echinochloa pyramidalis* and *Phragmites mauritianus* together with a ground sward of *Leersia hexandra*. *Vossia cuspidata* and *Aeschynomene nilotica* are also found in this group. In the mouths of two of the major rivers in the catchment (the Domasi and the Likangala), this group gives way to group A2b (Fig. 4). This is an association dominated by *Cyperus papyrus*. Because these stands are near rivers, they often have shallow standing water for all or most of the year. This group occurs on the most acidic soils of the Chilwa basin (pH below 5.0), but forms a very small part of the total vegetation. Groups A2c and A2d occur widely down the west shore. Both have *C. procerus* as the dominant sedge, and a dense ground sward of *Leersia hexandra*. Group A2d (average soil pH 5.9) is floristically richer than A2c (average soil pH 7.0), with the sedges *Cyperus articulatus* and *Scirpus maritimus* often mixed with the *Cyperus procerus*. These areas are generally dry for 150 to 200 days per year, and flood to a water depth of up to 50 cm.

Type A3: Intermediate. This single stand has species common to both A2 and A4 vegetation types and represents a transition between the two.

Type A4: Alkaline marsh. Alkaline marsh occurs at the southern end of Lake Chilwa on heavy clay soils with a pH exceeding 7.5. The vegetation consists of a mixed sward of *Vossia cuspidata* and *Cyperus longus*, interspersed with clumps of the robust legume *Aeschynomene pfundii*. *Typha domingensis* and *Scirpus maritimus* are also sometimes present.

Types A5 and B1: Disturbed areas. These two types, represented by single stands, although completely different in vegetation, are dealt with together for convenience as both were heavily disturbed by human activities. Type A5 is characteristic of the few sandy beaches around the lake where there is intensive seine net fishing. The grasses *Eriochloa borumensis* and *Paspalidium geminatum* predominate in a mixed sward. In type B1, in an area which had previously been cultivated for rice, *Nymphaea caerulea*, the wild rice *Oryza sativa* and the semi-floating *Utricularia reflexa* are the most abundant.

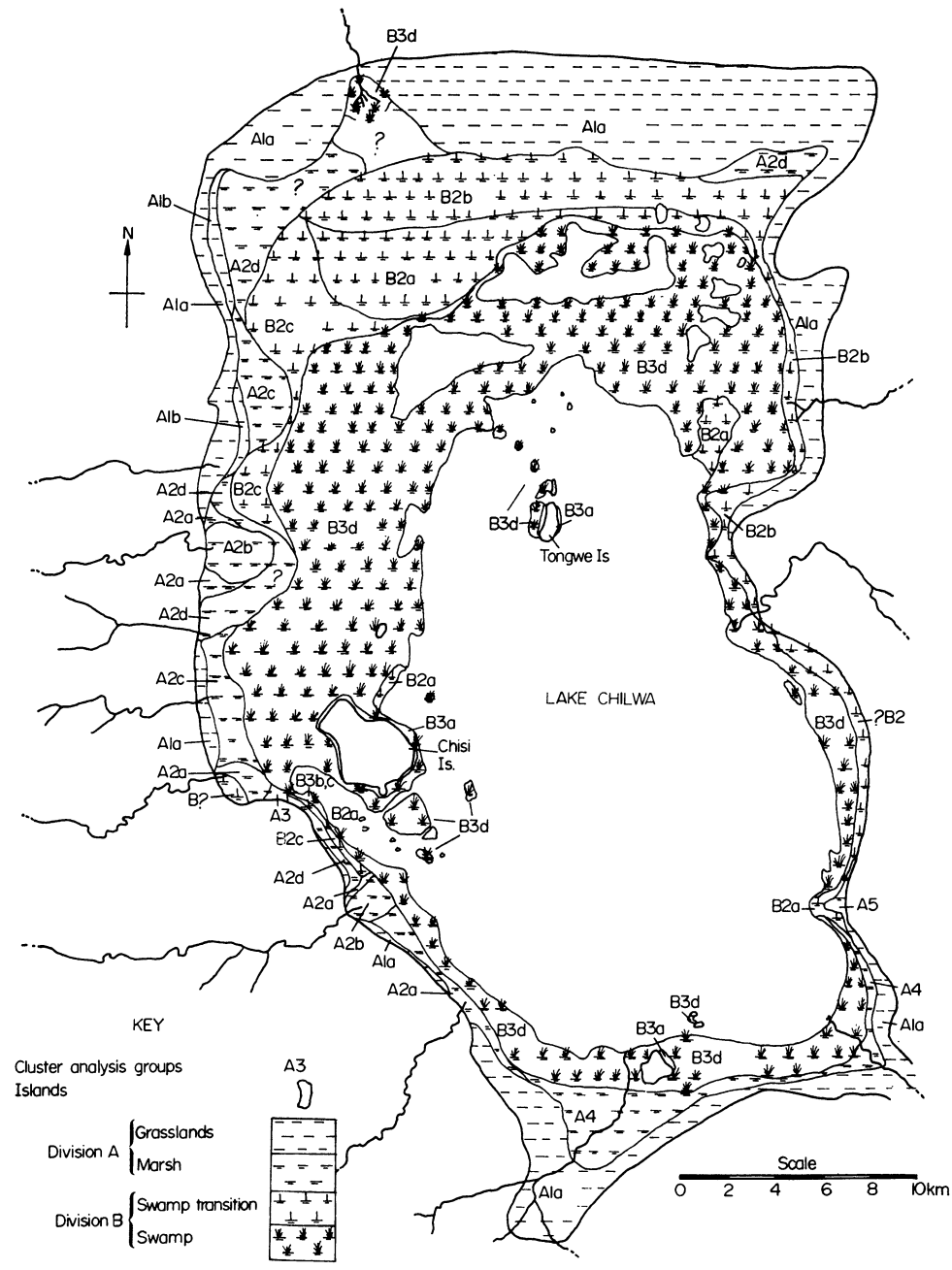


FIG. 4. Vegetation map of Lake Chilwa. The vegetation is divided according to the classification derived from the cluster analysis (Fig. 3).

Nymphaea occurs in all areas disturbed for rice cultivation, and is probably spread by wild ducks as the seeds are a major component of their diet (G. M. Schulten, unpublished), and paddy fields are a common duck feeding ground.

Type B2: Swamp transition. Fig. 4 shows that this vegetation type occupies extensive areas in the lake basin, particularly in the north and to some extent down the east side of the lake. It may also be called an alkaline grassland (cf. Vesey-Fitzgerald 1970) as the grasses *Diplachne fusca* and *Panicum repens* usually form most of the plant biomass. These areas are dry for 50 to 150 days each year, but are frequently flooded for short periods during the dry season when wind-generated surges of water from the lake (wind 'tides') flood the north plains. The soils are heavy alkaline clays. Group B2a is a simple vegetation type consisting of a sward of *Diplachne fusca* with very small circular patches of *Cyperus laevigatus* in hollows. Group B2b differs from this in that *Panicum repens* and *Scirpus maritimus* are mixed with *Diplachne fusca*. Group B2c, found on the west side, has the above species with locally common patches of *Vossia cuspidata*, *Cyperus* spp. and sometimes *Typha domingensis*.

Type B3: Swamp. Swamp forms the largest area of marginal vegetation around Lake Chilwa (Fig. 4), occurring on soils which are seldom dry, and are alkaline (pH around 8.5) and relatively saline (conductivity 3000 $\mu\text{mho cm}^{-1}$). Groups B3a, B3b and B3c are dominated by *Phragmites mauritianus* and occur on the edges of the lake islands. They separated on the dendrogram because of different species of floating plants occurring below the *Phragmites*. The stands of the largest group, B3d, are in deeper water in a swamp community dominated by *Typha domingensis* which forms most of the Lake Chilwa vegetation. Only subgroups of particular interest are considered below. A full description is given by Howard-Williams (1973).

Opposite perennial river mouths on the lake edges of the swamp are large floating swards of *Vossia cuspidata*, often 100 m across. *Pistia stratiotes* and *Utricularia* spp. are frequently present among the culms. Stands of this vegetation type fall into subgroup 2, and resemble closely the 'floating meadows' of the Amazon River system (Junk 1970, 1973).

Subgroup 4 consists of clumps of the grass *Paspalidium geminatum* in the open lake up to 5 km from the swamp. This grass can tolerate water up to 250 cm deep and vigorous wave action. Also found in large patches out in the lake south of Kachulu Bay and north of Tongwe Islands is the large sedge *Scirpus littoralis* (subgroup 5). This species normally inhabits coastal estuaries in south-east Africa (Breen & Jones 1970; Millard & Broekhuysen 1970) and Lake Chilwa is to date the only inland water body where this plant has been found.

Subgroup 6 comprises deep water stands with *Aeschynomene pfundii*, which is normally associated with alkaline marsh vegetation. This species invaded the mud flats on the south-west edge of the lake during the dry period of 1968 (Fig. 2). This particular vegetation took the form of a 'swamp forest' of *Aeschynomene pfundii* 4 to 5 m high, with *Nymphaea* spp. and *Ceratophyllum demersum* appearing at the water surface. The 'swamp forest' persisted for 2 to 3 years after the lake refilled, and then died off, as the *Aeschynomene* was unable to regenerate in the deep water of the lake.

Fig. 3 shows that subgroup 10 is a cluster of very highly correlated stands. These come from the centre of the Lake Chilwa swamps where the vegetation is usually a pure stand of *Typha domingensis*. *Nymphaea caerulea*, *Utricularia inflexa* and *Ipomoea aquatica* are sometimes found in disturbed clearings amongst the *Typha*.

Interpretation of the cluster analysis and vegetation map indicated that water depth and soil pH were the most important environmental variables affecting the Lake Chilwa vegetation. It was hoped that an ordination by principal components analysis would allow a clearer interpretation of the environmental gradients operating on and within the Lake Chilwa vegetation.

Ordination

The species loadings and the proportions of the total variance extracted from the first five components are given in Table 3. Almost 70% of the total variance is accounted for by the first five components. Since much of the variation in species composition in any

Table 3. *Principal components analysis: species loadings on the first five components*

Species	Symbol	Component				
		1	2	3	4	5
<i>Typha domingensis</i>	Typ	-0.632	-0.144	-0.010	0.246	-0.425
<i>Ceratophyllum demersum</i>	Cer	-0.595	-0.124	0.576	-0.141	0.049
<i>Pistia stratiotes</i>	Pis	-0.611	-0.057	0.511	-0.121	0.131
<i>Lemna perpusilla</i>	Lem	-0.515	0.470	0.125	0.044	0.162
<i>Spirodela polyrrhiza</i>	Spi	-0.343	0.054	0.520	0.017	0.242
<i>Ipomoea aquatica</i>	Ipo	-0.303	0.521	0.212	0.243	-0.055
<i>Panicum repens</i>	Pan	-0.297	-0.053	0.273	0.595	0.457
<i>Cyperus articulatus</i>	C. art	-0.270	0.006	0.219	0.486	0.540
<i>Utricularia inflexa</i>	U. in	-0.532	0.145	0.416	-0.135	0.229
<i>Diplachne fusca</i>	Dip	-0.713	-0.097	-0.026	0.340	0.259
<i>Paspalidium geminatum</i>	Pas	-0.546	-0.197	0.570	-0.016	-0.033
<i>Aeschynomene pfundii</i>	A. pf	-0.017	0.224	0.407	-0.160	-0.189
<i>Nymphaea caerulea</i>	Nym	-0.606	0.309	0.025	-0.026	0.212
<i>Vossia cuspidata</i>	Vos	-0.031	0.551	0.366	0.115	-0.226
<i>Scirpus littoralis</i>	S. lit	-0.595	-0.190	0.555	-0.202	-0.122
<i>Cyperus laevigatus</i>	C. la	-0.779	-0.047	-0.006	0.320	0.023
<i>Pseudowolfia hyalina</i>	Ps	-0.113	-0.016	0.633	0.003	0.089
<i>Cyperus alopecuroides</i>	C. alo	0.074	0.254	0.444	0.035	0.250
<i>C. procerus</i>	C. pr	0.454	0.231	-0.154	0.356	0.336
<i>C. digitatus</i>	C. dig	0.220	0.758	0.336	0.073	-0.118
<i>Utricularia reflexa</i>	U. re	-0.176	0.620	-0.140	0.294	0.265
<i>Alternanthera sessilis</i>	Alt	-0.202	0.675	0.075	0.227	0.276
<i>Phragmites mauritianus</i>	Phr	-0.015	0.583	0.134	-0.226	-0.046
<i>Cynodon dactylon</i>	Cyn	0.794	-0.349	0.254	0.102	0.034
<i>Commelina africana</i>	Com	0.107	0.366	0.301	-0.223	0.110
<i>Eriochloa borumensis</i>	Eri	0.603	0.175	0.427	0.220	-0.218
<i>Leersia hexandra</i>	Lee	0.503	0.712	-0.255	-0.055	-0.040
<i>Echinochloa pyramidalis</i>	Ech	0.318	0.701	-0.047	-0.274	0.082
<i>Aeschynomene nilotica</i>	A. nil	0.123	0.843	0.322	0.107	-0.029
<i>Wolffiopsis welwitschii</i>	Wol	-0.065	0.866	0.146	-0.030	0.208
<i>Oryza sativa</i>	Ory	0.016	0.785	-0.100	-0.265	0.257
<i>Scirpus maritimus</i>	S. mar	0.387	-0.119	-0.094	0.603	0.026
<i>Hyparrhenia rufa</i>	Hyp	0.857	-0.235	0.137	-0.095	0.063
<i>Eragrostis gangetica</i>	Era	0.832	0.098	0.229	0.111	0.064
<i>Hygrophila auriculata</i>	Hy	0.789	0.005	0.232	0.241	0.023
<i>Sporobolus pyramidalis</i>	Spo	0.870	-0.237	0.043	0.188	0.177
<i>Chloris virgata</i>	Chl	0.855	-0.126	0.325	0.155	-0.118
<i>Cyperus longus</i>	C. lon	0.526	0.056	0.119	0.400	-0.570
<i>Pycnus mundtii</i>	Pyc	0.350	0.149	0.100	0.434	-0.661
<i>Utricularia gibba</i>	U. gib	-0.215	0.504	-0.205	0.530	-0.053
<i>Cyperus papyrus</i>	C. pap	-0.105	0.697	-0.101	0.187	-0.431
<i>Polygonum limbatum</i>	Poly	-0.353	0.677	0.075	0.006	-0.225
Proportion of total variance (%)		31.5	18.2	7.9	6.2	5.3
Accumulated variance (%)		31.5	49.7	57.6	63.8	69.1

area is due to chance, components which extract less than 5% of the total variance are probably not worth examining.

The species with the highest positive loadings on the first component are *Sporobolus pyramidalis*, *Chloris virgata* and *Hyparrhenia rufa*, all plants of the marginal grasslands (Fig. 5). The species with the largest negative loadings include *Cyperus laevigatus*, *Diplachne fusca*, *Typha domingensis*, *Pistia stratiotes* and *Ceratophyllum demersum*. In 1970 when this study was conducted, these last species were inhabitants of the deepest waters on the lake edges of the swamp. The first component, accounting for 31.5% of the variation in the vegetation, seems to reflect a water depth gradient. However, it will

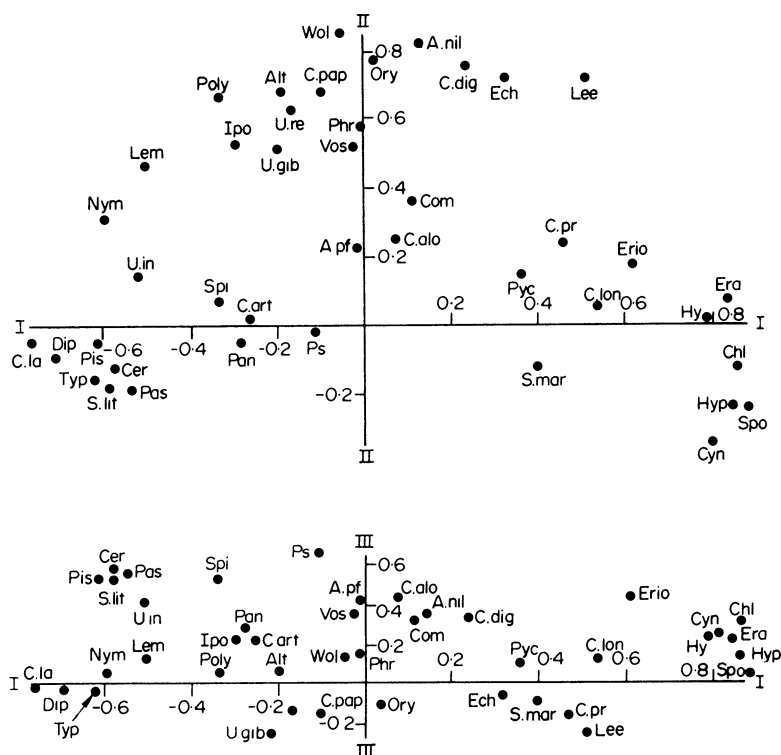


FIG. 5. Principal components analysis. Ordination of the adjusted species loadings on the first three axes. The lengths of the axes are proportional to the amount of total variance that they account for. The full name for each species abbreviation is given in Table 3.

be shown later that the interpretation of this component in the stand ordination rests on a number of other environmental factors as well. After this study, *Cyperus laevigatus* and *Diplachne fusca* died out in the lake, where they had grown for two to three years following colonization of the lake bed during the dry period.

Species with the highest positive loadings on the second component are *Wolffiopsis welwitschii*, *Aeschynomene nilotica*, *Oryza sativa*, *Leersia hexandra*, *Cyperus digitatus* and *C. papyrus*. These species are all found in relatively acidic soils (pH below 6.0), in the neutral to acidic marsh vegetation type. The grassland species *Cynodon dactylon*, *Hyparrhenia rufa*, *Sporobolus pyramidalis* and the deep water swamp species *Scirpus littoralis*, *Paspalidium geminatum* and *Typha domingensis* have the lowest negative loadings on the second component. All these plants grow on alkaline soil. The second

component is interpreted as reflecting a pH gradient, and it accounts for a further 18.2% of the variation.

On the third component, the floating plants *Ceratophyllum demersum* and *Pseudowolfia hyalina* and the deep water open lake plants *Scirpus littoralis* and *Paspalidium geminatum* have the highest positive loadings, together with the other free-floating species *Pistia stratiotes*, *Spirodela polyrhiza* and *Utricularia inflexa*. Those with the highest

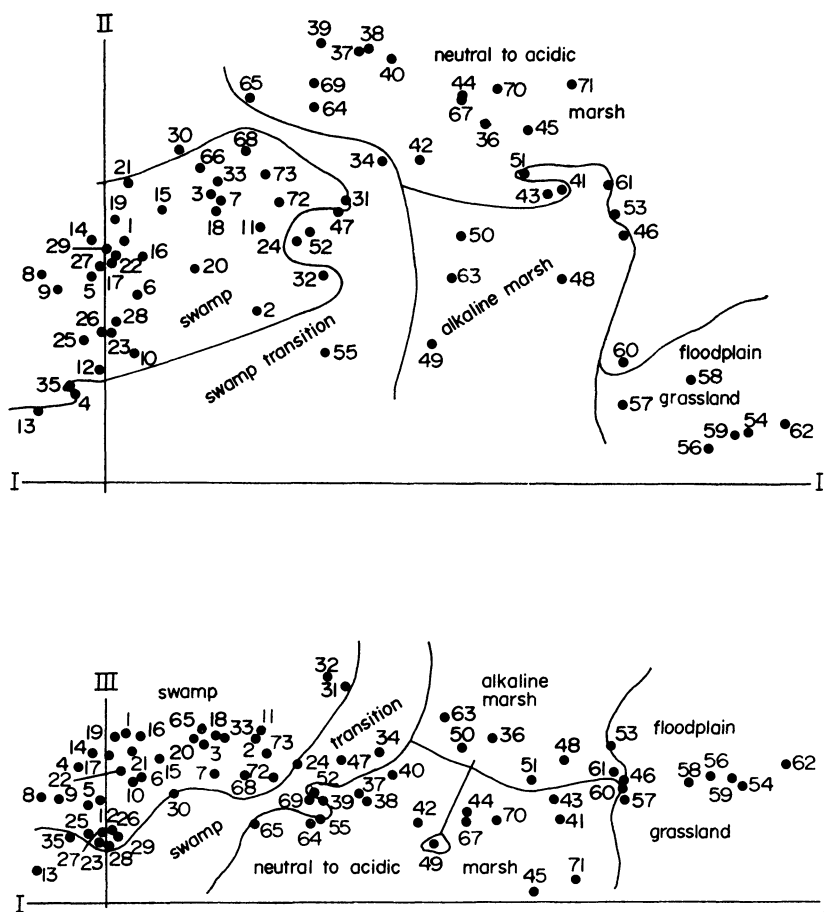


FIG. 6. Principal components analysis. Ordination of stands on the first three axes derived from a principal components analysis of the species data. The lengths of the axes are proportional to the amount of total variance that they account for. The classification of stands according to the cluster analysis has been superimposed on the ordination diagrams.

negative loadings are the marsh grass *Leersia hexandra* and the characteristic *Utricularia* species of shallow seasonal waters, *U. gibba* and *U. reflexa*. This component seems to account for residual variation in both water regime and pH gradients. No interpretation is offered for the other components.

The stand ordinations derived from the species data are shown in Figs. 6 and 7. The classifications derived from the cluster analysis are superimposed on the ordination diagrams in Fig. 6 and show a good agreement with the ordination. In Fig. 7 only the

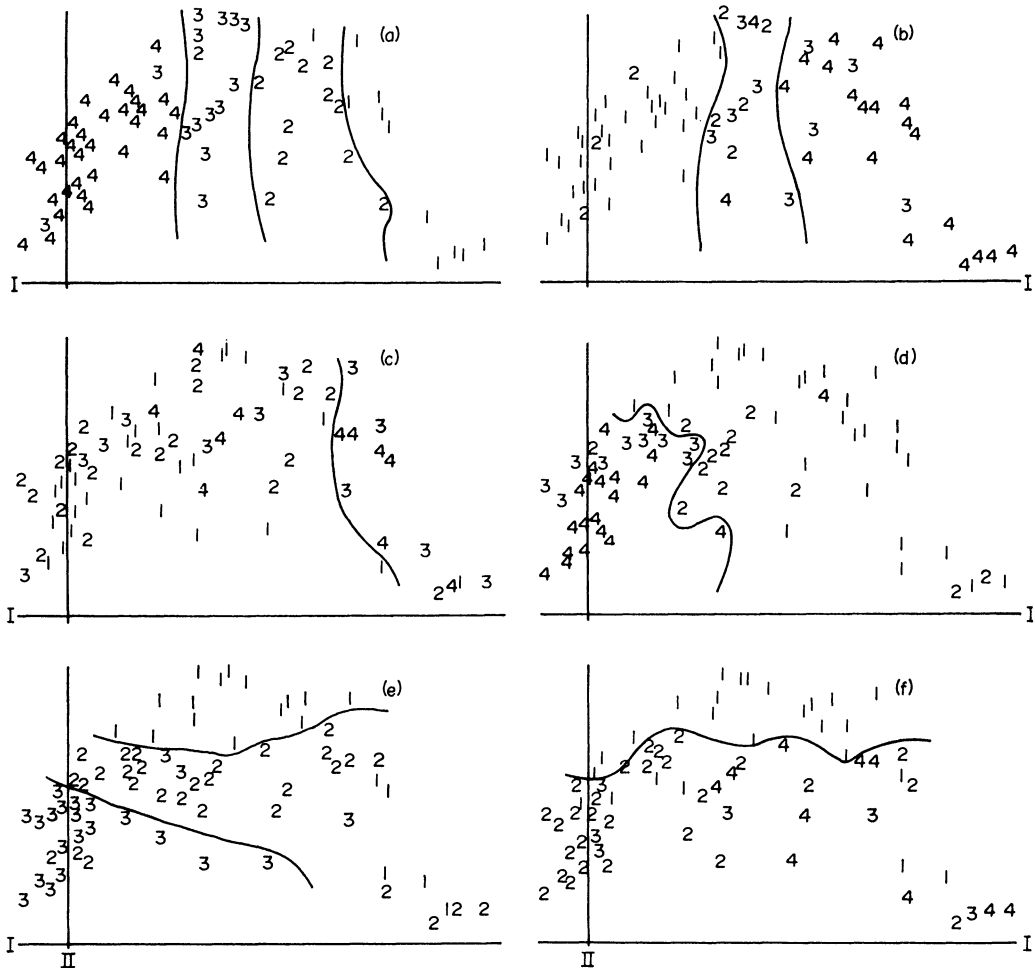


FIG. 7. Levels of selected environmental factors superimposed on the stand ordinations. The lengths of the axes are proportional to the amount of total variance that they account for. The first two axes only are shown. (a) Dry season water depth: 1, > 50 cm below the soil; 2, 20 to 49 cm below the soil; 3, 20 cm above to 19 cm below the soil; 4, > 20 cm above the soil. (b) Dryness (days per year): 1, < 50; 2, 50–150; 3, 150–250; 4, > 250. (c) Disturbance: see text for ratings; 4 is used in the figure for ratings 4, 5 and 6. (d) Soil conductivity ($\mu\text{mho cm}^{-1}$): 1, < 1000; 2, 1000–1999; 3, 2000–3000; 4, > 3000. (e) Soil pH: 1, < 7.0; 2, 7.0–8.0; 3, > 8.0. (f) Total soil phosphorus (mmol kg^{-1}): 1, < 1.6; 2, 1.6–3.2; 3, 3.2–4.8; 4, > 4.8.

interpretable environmental factors measured are superimposed on the stand ordination. As shown by Howard-Williams (1972) there are strong correlations between soil and water conductivity values ($P < 0.001$) and between soil and water pH values ($P < 0.05$). Nitrate and phosphate showed very little variation and water clarity was negatively correlated with water depth ($P < 0.001$), so water factors are not displayed on the ordination diagrams.

The first component, as was suggested from the species ordination, shows conspicuous arrangement of stands in relation to water depth and dryness (Fig. 7a, b) and, to some extent, disturbance (Fig. 7c). Disturbance and water depth are, however, negatively

correlated, as most human activities are in the drier areas. An ordination of stands on the first component in terms of soil conductivity (Fig. 7d) is also apparent, and there is some similarity between the soil conductivity distribution pattern and the distribution pattern of vegetation types on the stand ordination (Fig. 6). This could also indicate that soil salinity exerts some control on the vegetation types. The principal components analysis did not separate the effects of water depth and salinity, presumably because these two factors are related in the Chilwa basin. The correlation coefficient between water depth and soil conductivity was $r=0.49$, $P<0.001$.

The second component shows that soil pH, and to some extent soil phosphorus, influence the stand distribution. Surprisingly, no relationships between soil texture and vegetation distribution could be established on any of the first five components.

From interpretations of the cluster analysis and of the species and stand ordinations by principal components analysis, it seems that the major environmental controls on the Lake Chilwa vegetation are water regime and the associated soil salinity, disturbance, and soil pH.

DISCUSSION

First the Lake Chilwa vegetation is considered in relation to that of other swamps and marshes in East and Central Africa. There are two distinct types of large swamp in this part of Africa; those associated with endorheic regions and/or alkaline brackish regions, and those of exorheic regions and/or where the water conductivity is low. There are many endorheic areas in East Africa, often produced by tectonic movements associated with the Rift Valley, described for instance by Talling & Talling (1965). In these regions, weathering results in the accumulation of bases, mainly sodium carbonate, producing alkaline conditions in the waters and soils. In contrast, the exorheic drainage systems generally have a low ionic content (conductivity of less than $300 \mu\text{mho cm}^{-1}$) and lower pH (7.0 or below in most rivers). However, highly productive lakes e.g. Lake George (Royal Society African Freshwater Biological Team 1970) naturally have a high water pH. Swamps dominated by *Typha* spp. (usually *T. domingensis*) prevail in alkaline and endorheic areas (e.g. Lake Chilwa, Lake Manyara, Lake Upemba), whilst *Cyperus papyrus* swamps characterize exorheic, low-conductivity systems (e.g. River Nile, Lake Victoria, Lake Bangweulu). In both these swamp types the major control on the vegetation is water depth, and there is a distinct vegetation zonation from the deep water to the seasonally flooded marginal soils. The species composition not only of the swamps, but also of the marshes and grasslands, depends on the nature of the drainage system.

In exorheic or low-conductivity waters, *C. papyrus* inhabits the deepest waters of the swamp, often forming a floating sudd (Eggeling 1935; Carter 1955). *Melanthera scandens* and *Polygonum* spp. are frequent associates of the papyrus. The marshes behind these swamps are normally occupied by robust grasses such as *Vossia cuspidata*, *Echinochloa pyramidalis*, *Miscanthidium violaceum* and *Phragmites mauritianus*. *Leersia hexandra*, *Loudetia phragmitoides*, and many Cyperaceae, particularly *Eleocharis* spp. and *Cyperus* spp., occur in shallower waters in the marshes. The floodplain grasses in this area consist largely of *Andropogon* spp., *Hyparrhenia* spp. and *Loudetia simplex* (Lind 1956; Greenway & Vesey-Fitzgerald 1969; Vesey-Fitzgerald 1970).

In contrast to this, the wetland vegetation on alkaline soils and around many endorheic water bodies (conductivity up to $12\,000 \mu\text{mho cm}^{-1}$) is characterized by a *Typha* community in the deepest waters of the swamps. The sedges *Cyperus laevigatus* and *Scirpus*

maritimus are common in the marshes behind such swamps, whilst the grasses *Diplachne fusca*, *Sporobolus spicatus*, *S. consimilis*, *Chloris gayana* and *Cynodon dactylon* are common on East African alkaline floodplains.

It appears therefore that the two major types of swamp in East and Central Africa are differentiated on the basis of pH and salinity. *Cyperus papyrus* can tolerate a wide range of environmental conditions, but according to Lind (1956) it is restricted to a pH range of 5.5 to 6.4. In Lake Chad, *C. papyrus* is generally replaced by *Phragmites* when the water conductivity exceeds $600 \mu\text{mho cm}^{-1}$ (J. P. Carmouze, personal communication). In the swamps of Inhaca Island, Moçambique, Noel (1959) found that *Cyperus papyrus* occurred only in the most acidic swamps investigated (pH 4.6). The other swamps were dominated by *Typha australis* (= *T. domingensis*). Data produced in this paper show that, in Lake Chilwa, *Cyperus papyrus* is confined to two small areas with acidic soils (pH 4.5) and low conductivity (30 to $100 \mu\text{mho cm}^{-1}$), whereas the remaining vegetation is typical of alkaline brackish areas.

In the PCA used in this study, only 47% of the variation in the vegetation (accounted for by the first two components) could be directly related to the measured environment. This low value can be partly attributed to the distortions arising from non-linear data (Austin & Noy-Meir 1971). Nevertheless there is a considerable proportion of unexplained variance. Furthermore, in contrast to the findings of many authors (e.g. Pearsall 1920; Curtis 1959; Spence 1964; Denny 1972; Seddon 1972) soil texture did not seem to play an important role in determining vegetation types in Lake Chilwa. Unlike many other African lakes where there is a distinct zone of submerged or floating-leaved plants (Eggeling 1935; Carter 1955; Walter 1971) or a floating sudd (Lind & Visser 1962; Mitchell 1969) in deep water on the lake side of the swamps, the stands of *Typha* sharply delimit the lake-swamp boundary of Lake Chilwa. There are no submerged species in the open lake, and distinct zones of floating-leaved plants are confined to small sheltered inlets in the *Typha*.

In order to explain these anomalies and to discuss the composition of, and patterns within, the Lake Chilwa vegetation it is necessary to examine the different kinds of pressure which cause vegetation change. According to Hutchinson (1953) four interacting factors cause variation in vegetation within an area. These are (i) vectorial: due to external gradients, e.g. water depth, salinity; (ii) coactive: interaction of species through competition; (iii) reproductive: growth of offspring around the parent; (iv) stochastic: random and other forces unrelated to the system.

The absence of submerged or floating-leaved plant communities in Lake Chilwa can be explained entirely by vectorial forces. The high turbidity (Howard-Williams 1972), which prevents light from penetrating further than 10 to 12 cm, and the turbulence of the shallow lake waters prevent the growth of submerged plants. Wave action also inhibits the development of communities of floating-leaved plants in all but sheltered bays. Vectorial forces (high pH, conductivity and wave action) also prevent colonization of the lake by sudd-forming species such as *Cyperus papyrus*, *Vossia cuspidata* and *Echinochloa pyramidalis*.

High soil pH favours the establishment of the dominant swamp plant, *Typha domingensis*, and once established it creates its own local environmental conditions. Within the *Typha* swamp, there is a lack of space, no wave action except on the lake edges, surface light levels are greatly reduced, and the chemistry of the waters alters considerably (Howard-Williams 1972). The presence of *Typha* therefore controls the remaining species composition of the swamp, and any variation in species composition will be due to

changes in *Typha* itself. The Lake Chilwa *Typha* swamp is therefore structurally very simple and often consists of pure stands. This floristic simplicity in a swamp community is a widespread phenomenon (Curtis 1959; Sculthorpe 1967; Walker 1968; Szczepanska 1971) and has been ascribed to the vigorous vegetative reproductive capacity of many emergent macrophyte species. Variation in the vegetation within such a swamp is therefore dependent to a large extent on chance (stochastic factors). The presence of floating species, such as *Pistia stratiotes*, *Lemna perpusilla* and *Spirodela polyrhiza* on the lake edges of the Lake Chilwa swamps is a result of their chance transportation there by wind, and not of a favourable physical or chemical environment. The presence of plants from the marshes and swamp transition areas on the seasonally dry landward edges is a result of chance germination of seeds in more open patches of *Typha*. Variation in the centre of the swamp occurs in small lagoons where the *Typha* has been cut by fishermen to allow the placement of gill nets and fish traps. *Nymphaea caerulea* and *Utricularia inflexa* are common here, and the twining *Ipomoea aquatica* and the submerged *Ottelia ulvifolia* exploit the increased light.

In summary, therefore, change in species composition within the basically uniform *Typha* swamp community is not a direct result of the measured environmental factors, but is to a large extent due to stochastic factors. This, and the fact that the swamp itself is floristically very poor, must be included in the interpretation of vegetation ordinations. If stands are located in an area which has only three or four species, the absence of one of the species or the presence of an additional one may result in a considerable interstand distance, whereas in a community with, say, twenty species per stand, a difference of five or six species may be required to produce a similar interstand distance. There is a high probability that the presence of one new species in a stand is due to chance, but the presence of five or six new species in a stand is more likely to be the result of an environmental difference. Therefore, in vegetation types with a few species, the stochastic element plays a far greater role in determining relative positions and interstand distances in an ordination than in a floristically rich community where similar interstand distances can be interpreted in terms of an environmental difference.

The change in vegetation on the landward edges of the *Typha* is a response to water depth. On alkaline soils, the *Typha* swamp is replaced by a swamp transition vegetation type. Along the west shore, the hygrophilous grasses *Panicum repens* and *Vossia cuspidata*, and the sedges *Scirpus maritimus*, *Cyperus procerus* and *C. articulatus* occur. In the north, however (Fig. 3), where wind-generated surges of water from the lake keep the soils moist for most of the year, *Diplachne fusca* and *Cyperus laevigatus* are widespread. As the water depth increases further, a marsh vegetation develops. This is extremely variable, the major variations being related to soil pH differences. On the driest soils of the floodplain, marsh is replaced by floodplain grassland.

In the swamp transition with marsh and floodplain grassland vegetation types, the influence of water depth on the vegetation is greater than in the swamp where the water is deeper. This is because in areas of shallow water, small fluctuations cause large variations in species composition whereas similar fluctuations would have little effect in the deeper swamp waters. This was clearly demonstrated by Walker & Coupland (1968). Intrusions of drainage lines, which can be considered a vectorial force, therefore cause considerable variation in the vegetation on the landward edges of the swamp. These variations, together with local variations caused by differential burning and grazing and human disturbance in the dry season, result in mosaic patterns in the vegetation. Van Rensburg (1971) noted similar mosaics resulting from burning in the Kafue River floodplain in

Zambia. This mosaic effect is enhanced in some areas subjected to equal wet and dry periods, which results in different species having the competitive advantage in each season. This means that stands with more than one dominant species result.

The fact that soil texture did not have a marked effect on the vegetation is probably due to the nature of the Lake Chilwa plant communities. Studies in which soil texture has proved to be of importance (see references on p. 849) dealt mainly with submerged, rooted aquatic plants. These do not occur in Lake Chilwa, and the vigorous rhizomatous nature of the emergent plants here means that their resistance to wave action is probably equal on sand and clay. The nutrient-rich water of the lake and swamp (McLachlan *et al.* 1972; Howard-Williams 1972) which easily percolates into sandy soils probably counteracts the other effect of a sandy substrate, that of possessing a low nutrient content and exchange capacity.

When this study was carried out, Lake Chilwa had just refilled after a dry period. This dry period affected the vegetation of the lake in that three species, *Aeschynomene pfundii*, *Diplachne fusca* and *Cyperus laevigatus* which normally inhabit the drier marsh and swamp transition vegetation types, were able to invade the lake bed. They were consequently found in deep water for two to three years after the lake refilled (C. Howard-Williams, unpublished). In the present context, therefore, the dry period exerted a strong stochastic effect on the vegetation, and rendered the ordination less effective because common species of the dry areas were found together with species of the deeper waters. A knowledge of recent past events in the environment is thus an essential part of the correct interpretation of an ordination.

About half the variation in the vegetation around Lake Chilwa can plausibly be accounted for in terms of the measured environment. Water depth, and soil pH and salinity are important vectorial forces. A loss in sensitivity in the ordination due to low numbers of species in the swamp stands and the effect of a recent dry period in the lake basin may account to some extent for the apparently poor correlation between vegetation and environment, but clearly, in wetland vegetation, chance variation is extremely important. It can be demonstrated, therefore, that the divisions and vegetation types of the cluster analysis correspond to differences in environment, but that in many cases differences within the vegetation types are due largely to chance.

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SUMMARY

Vegetation and selected environmental factors were investigated for 73 stands in the marginal areas of Lake Chilwa, Malawi. A cluster analysis was used to classify the vegetation, and the vegetation groups so derived were used to construct a vegetation map of

Lake Chilwa. Principal components analysis was carried out on the species data to determine the relationships between species, and to ordinate the stands. By use of both these techniques, it appeared that a water depth gradient was the major controlling influence in the vegetation, although it was pointed out that this gradient was also correlated with salinity and disturbance. Soil pH and soil phosphorus accounted for less of the variation. Differences in soil texture did not seem to affect the vegetation, although chance appeared to play a very important role in determining the species composition of any stand. Most of the Lake Chilwa vegetation is made up of *Typha* swamp, and the relationship between this type of swamp and other East and Central African swamps is discussed.

REFERENCES

- Agnew, S. (1970). The people. A demographic study. *Decline and Recovery of a Lake* (Ed. by M. Kalk), pp. 41–4. Government Printer, Zomba.
- Austin, M. P. & Noy-Meir, I. (1971). The problem of non-linearity in ordination: experiments with two-gradient models. *J. Ecol.* **59**, 763–73.
- Beadle, L. C. & Lind, E. M. (1960). Research on the swamps of Uganda. *Uganda J.* **24**, 84–7.
- Bloomfield, K. (1965). The geology of the Zomba area. *Geol. Survey Dep. Bull.* No. 16. Government Printer, Zomba.
- Bray, R. H. & Kurtz, L. T. (1945). Determination of total and available forms of phosphorus in soil. *Soil Sci.* **56**, 39–45.
- Breen, C. M. & Jones, I. D. (1970). A preliminary list of Angiosperms collected in the vicinity of Lake Sibayi. *Trans. R. Soc. S. Afr.* **39**, 235–45.
- Carter, G. S. (1955). *The Papyrus Swamps of Uganda*. Heffer, Cambridge.
- Cook, C. D. K. (1968). The vegetation of the Kainji Reservoir site in Northern Nigeria. *Vegetatio*, **15**, 225–43.
- Curtis, J. T. (1959). *The Vegetation of Wisconsin*. University of Wisconsin Press, Madison.
- Curtis, J. T. & McIntosh, R. P. (1950). The interrelations of certain analytic and synthetic phytosociological characters. *Ecology*, **31**, 434–55.
- Denny, P. (1972). Sites of nutrient absorption in aquatic macrophytes. *J. Ecol.* **60**, 819–29.
- Drummond, H. (1902). *Tropical Africa*. Hodder & Stoughton, London.
- Drummond, R. B. (1972). A list of Rhodesian legumes. *Kirkia*, **8**, 209–29.
- Eggeling, W. J. (1935). The vegetation of Namnave Swamp, Uganda. *J. Ecol.* **23**, 422–35.
- Exell, A. W., Fernandes, A. & Wild, H. (1963). *Flora Zambesiaca*, Vol. 2. Crown Agents for Overseas Governments and Administrations, London.
- Exell, A. W. & Wild, H. (1960). *Flora Zambesiaca*, Vol. 1. Crown Agents for Overseas Governments and Administrations, London.
- Garson, M. S. (1960). The geology of the Lake Chilwa area. *Geol. Survey Dep. Bull.* No. 12. Government Printer, Zomba.
- Greenway, P. J. (1973). A classification of the vegetation of East Africa. *Kirkia*, **9**, 1–68.
- Greenway, P. J. & Vesey-Fitzgerald, D. F. (1969). The vegetation of Lake Manyara National Park. *J. Ecol.* **57**, 127–49.
- Howard-Williams, C. (1971). An ecological sampling method for swamp and marsh vegetation types. *News Lett. Limnol. Soc. South Afr.* **17**, 27–30.
- Howard-Williams, C. (1972). Limnological studies in an African swamp: seasonal and spatial changes in the swamps of Lake Chilwa, Malawi. *Arch. Hydrobiol.* **70**, 379–91.
- Howard-Williams, C. (1973). *Vegetation and environment in the marginal areas of a tropical African lake (Lake Chilwa, Malawi)*. Ph.D. thesis, University of London.
- Hutchinson, G. E. (1953). The concept of pattern in ecology. *Proc. Acad. nat. Sci. Philad.* **105**, 1–12.
- Hyder, D. N., Conrad, C. E., Tueller, P. T., Calvin, L. D., Poulton, C. E. & Sneva, F. A. (1963). Frequency sampling in sagebrush–bunchgrass vegetation. *Ecology*, **44**, 740–6.
- Jackson, M. L. (1958). *Soil Chemical Analysis*. Prentice-Hall, Englewood Cliffs, New Jersey.
- Junk, W. (1970). Investigations on the ecology and production biology of the floating meadows (*Paspalo-Echinochloetum*) on the Middle Amazon. 1. The floating vegetation and its ecology. *Amazoniana, Kiel*, **2**, 449–95.
- Junk, W. (1973). Investigations on the ecology and production biology of the floating meadows (*Paspalo-Echinochloetum*) on the Middle Amazon. 2. The aquatic fauna in the root zone of floating vegetation. *Amazoniana, Kiel*, **4**, 9–102.
- Kalk, M. (Ed.) (1970). *Decline and Recovery of a Lake*. Government Printer, Zomba.
- Lind, E. M. (1956). Studies in Uganda swamps. *Uganda J.* **20**, 166–76.

- Lind, E. M. & Visser, S. A. (1962). A study of a swamp at the north end of Lake Victoria. *J. Ecol.* **50**, 599–613.
- Marlier, G. (1973). Limnology of the Congo and Amazon Rivers. *Tropical Forest Ecosystems in Africa and South America; a Comparative Review* (Ed. by B. J. Meggers, E. S. Ayensu & W. D. Duckworth), pp. 223–38. Smithsonian Institution Press, Washington.
- McLachlan, A. J. & McLachlan, S. M. (1969). The bottom fauna and sediments in a drying phase of a saline African lake (L. Chilwa, Malawi). *Hydrobiologia*, **34**, 401–13.
- McLachlan, A. J., Morgan, P. R., Howard-Williams, C., McLachlan, S. M. & Bourn, D. (1972). Aspects of the recovery of a saline African lake following a dry period. *Arch. Hydrobiol.* **70**, 325–40.
- Millard, N. A. H. & Broekhuysen, G. J. (1970). The ecology of South African estuaries. 10: St. Lucia, a second report. *Zoologica Africana*, **5**, 277–307.
- Mitchell, D. S. (1969). The ecology of vascular hydrophytes on Lake Kariba. *Hydrobiologia*, **34**, 448–64.
- Morgan, A. (1968). *A physico-chemical study of Lake Chilwa*. M.Sc. thesis, University of Malawi.
- Morgan, A. & Kalk, M. (1970). Seasonal changes in the waters of Lake Chilwa (Malawi) in a drying phase 1966–1968. *Hydrobiologia*, **36**, 81–103.
- Morgan, P. R. (1972). Causes of mortality in endemic *Tilapia* of Lake Chilwa. *Hydrobiologia*, **40**, 101–19.
- Moss, B. & Moss, J. (1969). Aspects of the limnology of an endorheic African lake (L. Chilwa, Malawi). *Ecology*, **50**, 109–18.
- Müller, R. R. & Wideman, O. (1965). Bestimmung des Nitrat-Ions. *Vom Wass.* **22**, 247.
- Murphy, J. & Riley, J. P. (1958). A single solution method for the determination of soluble phosphate in sea water. *J. mar. biol. Ass. U.K.* **37**, 9–14.
- Noel, A. R. A. (1959). The vegetation of the fresh water swamps of Inhaca Island. *Jl S. Afr. Bot.* **25**, 189–205.
- Pearsall, W. H. (1920). The aquatic vegetation of the English lakes. *J. Ecol.* **8**, 163–201.
- Pike, J. G. & Rimmington, G. T. (1965). *Malawi: a Geographical Study*. Oxford University Press, London.
- Rensburg, H. J. van (1971). Fire: its effect on grasslands including swamps. *Proc. Tall Timbers Fire Ecology Conference*, **11**, 175–99. Tall Timbers Research Station, Tallahassee, Florida, U.S.A.
- Rice, E. L. (1967). A statistical method for determining quadrat size and adequacy of sampling. *Ecology*, **48**, 1047–9.
- Royal Society African Freshwater Biological Team (1970). Lake George, Uganda. *Studies on a Tropical Freshwater Ecosystem. Proc. IBP-UNESCO Symposium on Productivity Problems in Fresh Waters*, pp. 301–9. Kazimierz Dolny, Poland.
- Sculthorpe, C. D. (1967). *The Biology of Aquatic Vascular Plants*. Edward Arnold, London.
- Seddon, B. (1972). Aquatic macrophytes as limnological indicators. *Freshwat. Biol.* **2**, 107–30.
- Simon, B. K. (1971). Rhodesian and Zambian grass lists. *Kirkia*, **8**, 3–84.
- Spence, D. H. N. (1964). The macrophyte vegetation of freshwater lochs, swamps and associated fens. *The Vegetation of Scotland* (Ed. by J. H. Burnett), pp. 306–425. Oliver & Boyd, Edinburgh.
- Swan, J. M. A. (1970). An examination of some ordination problems by use of simulated vegetational data. *Ecology*, **51**, 89–102.
- Szczepanska, W. (1971). Allelopathy among aquatic plants. *Polskie Archiwum Hydrobiol.* **18**, 17–30.
- Talling, J. F. & Talling, I. B. (1965). The chemical composition of African lake waters. *Int. Revue ges. Hydrobiol. Hydrogr.* **50**, 421–63.
- Ungar, I. A. (1965). An ecological study of the vegetation of the big salt marsh, Stafford County, Kansas. *University of Kansas Science Bulletin*, **46**, 1–99.
- Vesey-Fitzgerald, D. F. (1970). The origin and distribution of valley grasslands in East Africa. *J. Ecol.* **58**, 51–75.
- Walker, B. H. (1968). *Ecology of herbaceous wetland vegetation in the aspen grove and grassland regions of Saskatchewan*. Ph.D. thesis, University of Saskatchewan.
- Walker, B. H. (1974). Some problems arising from the preliminary manipulation of plant ecological data for subsequent numerical analysis. *Jl S. Afr. Bot.* **40**, 1–13.
- Walker, B. H. & Coupland, R. T. (1968). An analysis of vegetation–environment relationships in Saskatchewan sloughs. *Can. J. Bot.* **46**, 509–22.
- Walker, B. H. & Coupland, R. T. (1970). Herbaceous wetland vegetation in the aspen grove and grassland regions of Saskatchewan. *Can. J. Bot.* **48**, 1861–78.
- Walker, B. H. & Wehrhahn, C. F. (1971). Relationships between derived vegetation gradients and measured environmental variables in Saskatchewan wetlands. *Ecology*, **52**, 85–95.
- Walkley, A. (1947). A critical examination of a rapid method for determining organic carbon in soils—effect of variations in digestion conditions and of inorganic soil constituents. *Soil Sci.* **63**, 251–64.
- Walter, H. (1971). *Ecology of Tropical and Sub-tropical Vegetation*. Oliver & Boyd, Edinburgh.
- Wild, H. (1961). Harmful aquatic plants in Africa and Madagascar. *Kirkia*, **2**, 1–64.

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APPENDIX 1

Authorities for species names mentioned in the text but not given in Table 2:

Chloris gayana Kunth

Loudetia phragmitoides (Peter) Hubbard

L. simplex (Nees) Hubbard

Melanthera scandens (Schumach. & Thonn.) Roberty

Miscanthidium violaceum (Schumach.) Robyns

Ottelia ulvifolia (Planch.) Walp.

Sporobolus consimilis Fresen

S. spicatus Kunth